

COAGULATION OF BLOOD

- **Clotting is the process** in which blood loses its fluidity and becomes a jelly-like mass few minutes after it is shed out.

FACTORS INVOLVED IN BLOOD CLOTTING

- Coagulation of blood occurs through a series of reactions due to activation of a group of substances called **clotting factors**.

Mnemonic for Clotting Factors

"Foolish People Try Climbing Long Slopes After Christmas, Some People Have Fallen"

Factor	First Letter	Name of Factor
Factor I	F	Fibrinogen
Factor II	P	Prothrombin
Factor III	T	Thromboplastin
Factor IV	C	Calcium (Ca ²⁺)
Factor V	L	Labile factor
Factor VII	S	Stable factor
Factor VIII	A	Antihemophilic factor A
Factor IX	C	Christmas factor
Factor X	S	Stuart–Prower factor
Factor XI	P	Plasma thromboplastin antecedent
Factor XII	H	Hageman factor
Factor XIII	F	Fibrin-stabilising factor

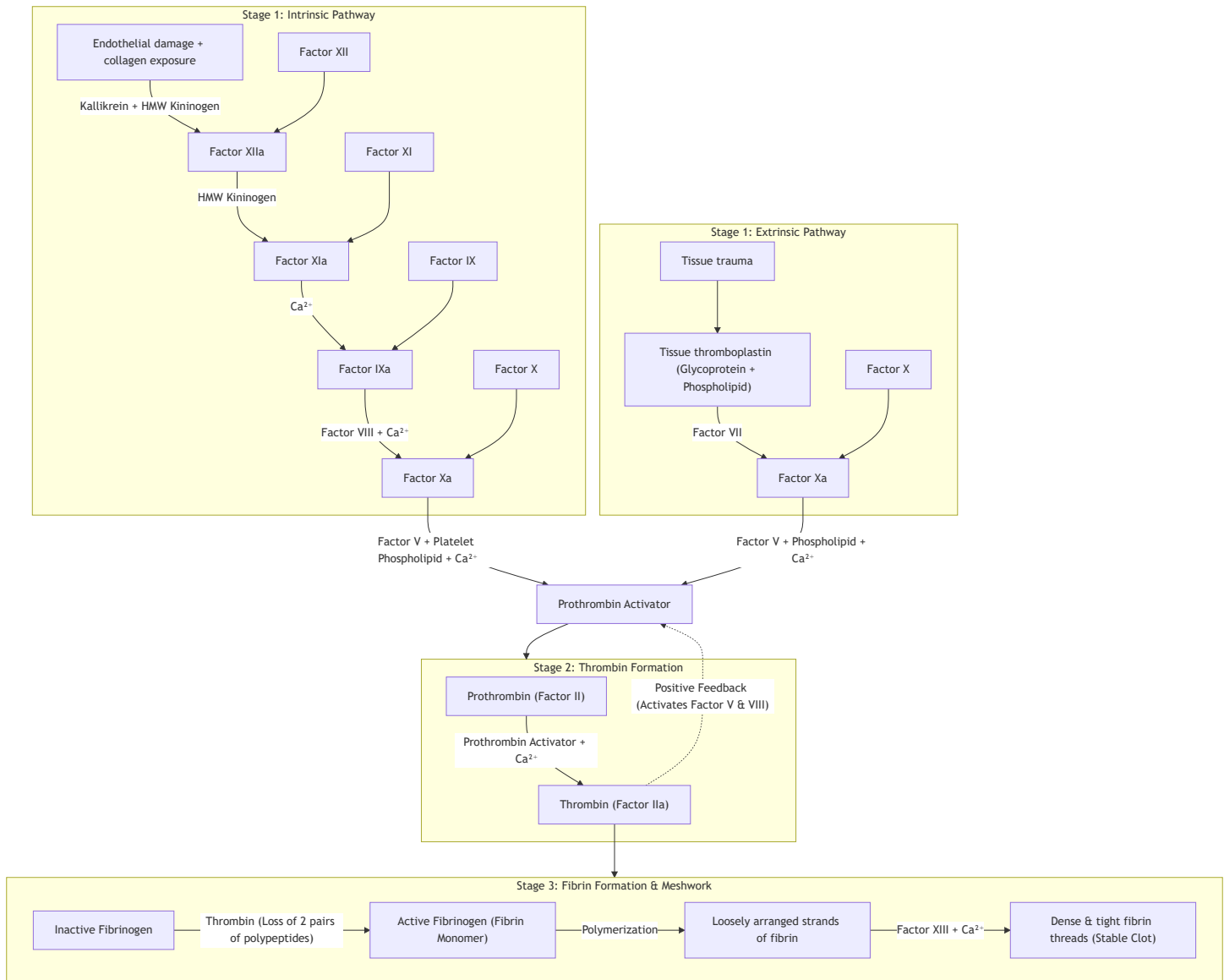
CLOTTING MECHANISM (ENZYME CASCADE THEORY)

- Clotting factors are proteins (*in form of enzymes*).
- All factors are present in the form of **inactive proenzymes** → converted into **active enzymes** → enforce clot formation.
- **Cascade Process** : Occurs through a series of steps, each step initiating the next until the final step is reached.

3 STAGES OF BLOOD CLOTTING

- a. **Stage 1** : Formation of prothrombin activator
- b. **Stage 2** : Conversion of prothrombin to thrombin
- c. **Stage 3** : Conversion of fibrinogen to fibrin

COMPREHENSIVE CASCADE (STAGES 1, 2 & 3)



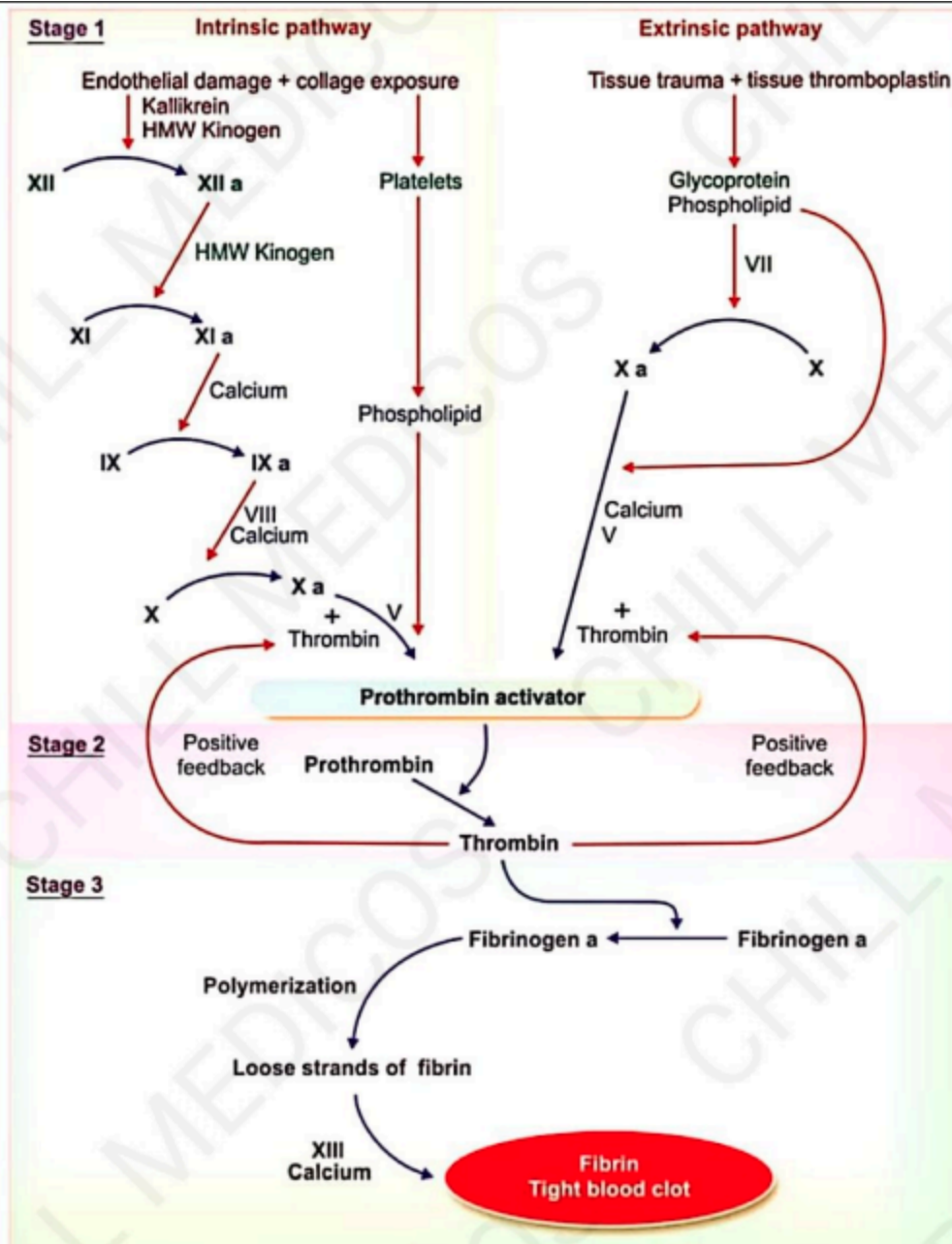


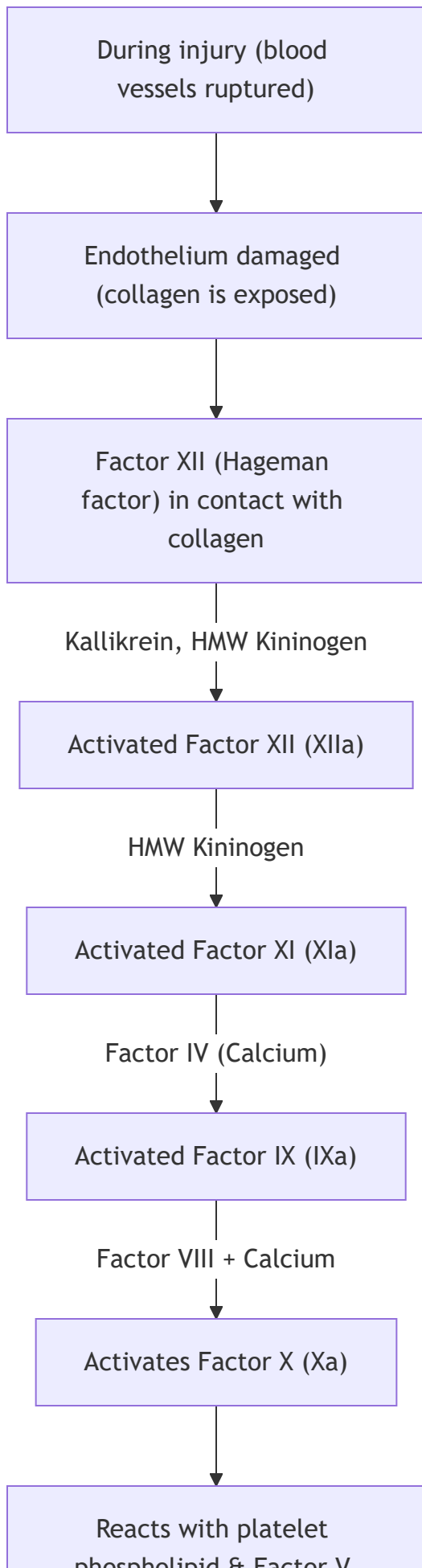
FIGURE 20.1: Stages of blood coagulation. a = Activated, + = Thrombin induces formation of more thrombin (positive feedback); HMW = High molecular weight.

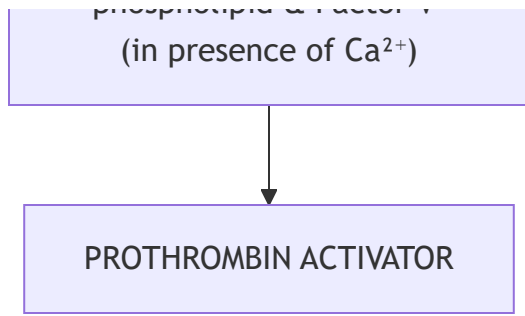
STAGE 1: FORMATION OF PROTHROMBIN ACTIVATOR

Occurs through **two pathways**:

1. Intrinsic Pathway

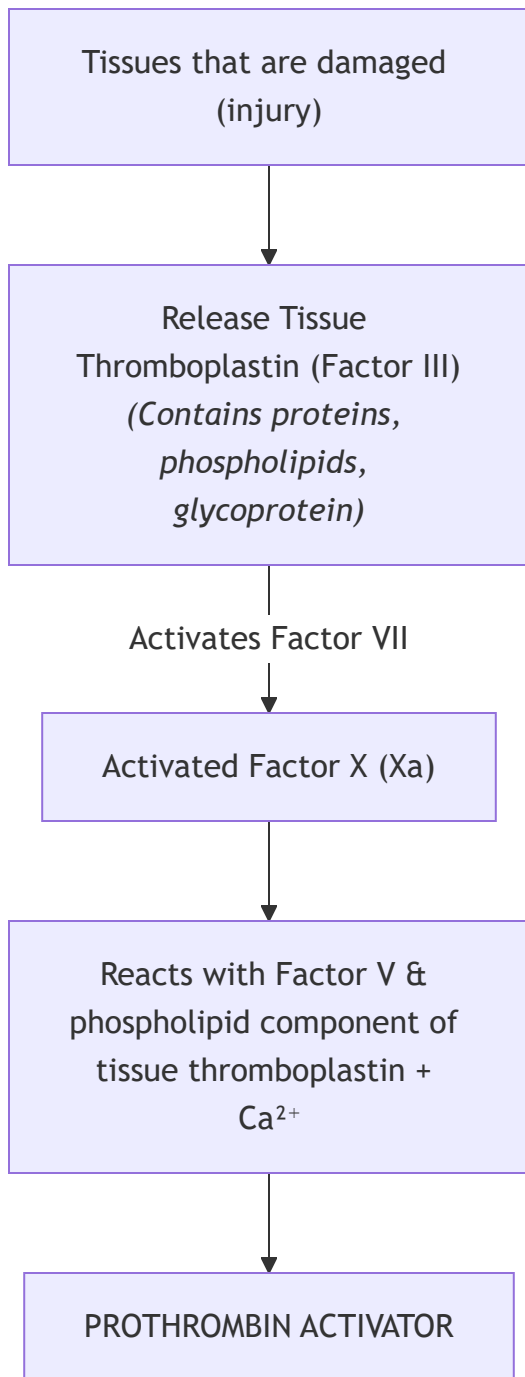
- Initiated by platelets **within blood itself**.





2. Extrinsic Pathway

- Initiated by **tissue thromboplastin** formed from injured tissues.



STAGE 2: CONVERSION OF PROTHROMBIN TO THROMBIN

- Blood clotting is centered around **thrombin formation**, which triggers clot assembly.

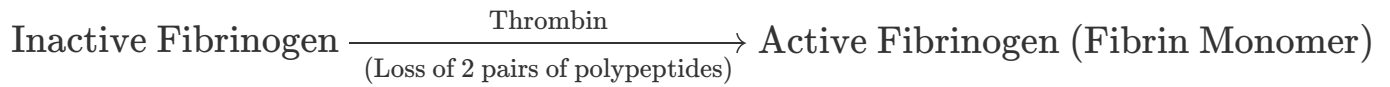


- **Positive Feedback Effect :—**

Thrombin accelerates the activation of **Factor V and Factor VIII**, accelerating further prothrombin activator formation.

STAGE 3: CONVERSION OF FIBRINOGEN TO FIBRIN

- Final stage of blood clotting.



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Polymerization

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Loosely arranged strands of Fibrin

↓ Factor XIII (Fibrin-Stabilizing Factor) + Ca^{2+}

Dense & tight fibrin threads

↓

Aggregate to form a meshwork of **Stable Clot**